



Electrochemical and spectroscopic studies of polymethacrylonitrile based electrolytes

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Abstract

Polymethacrylonitrile (PMAN)-based electrolytes were recently proposed as polymer matrix for ambient temperature applications. The living anionic polymerization of methacrylonitrile allowed PMAN–POE–PMAN triblock copolymers to be prepared from different starting PEGs. In order to appreciate the salt/polymer interactions, solvent-free polymer electrolytes were prepared by dissolution of lithium imide (LiTFSI) in PMAN homopolymer and copolymers. Despite the high T_g of PMAN host polymer, conductivities as high as 10^{-4} S cm $^{-1}$ were obtained around 90°C. Infra-red spectroscopy allowed the nitrile/salt interaction to be characterized. In particular, PMAN solvation number was determined from the homopolymer complexes. The investigation on the triblock copolymer complexes allowed the solvating competition between nitrile and ether functions to be highlighted. © 2000 Elsevier Science Ltd. All rights reserved.

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1. Introduction

The market of secondary batteries is indisputably pulled along by the explosion of portable electronics, namely the 3Cs: camcorders, computers and cellular phones. Thus, in 1996 the world production of secondary batteries represents about 150 million cells [1]. The market shares of secondary lithium batteries has been growing at a high rate since the emergence of commercial lithium-ion batteries around 1992, and these batteries are expected to have a prevalent position with the respect to alkaline batteries NiCd and NiMH from the end of the century. Besides the electronic applications, another market may appear for large capacity batteries, utility power sources (UPS) and electric vehicles (EV). Indeed high capacity batteries are needed not only in pure EV, but also in hybrid cars

which combine a thermal engine with an electrical one. Of course the battery specifications are markedly different in electrical and hybrid cars and at this moment several batteries may compete for these applications. Nevertheless, batteries are not the only solution for these applications and fuel cells, in particular PEMFC, present a decided advantage versus batteries in terms of autonomy and rechargeability. The PEMFC volume, their price, i.e. membrane and electrodes, still however, slow down their industrial development. Polymeric materials are used in all these systems: polyether based host polymers for lithium polymer batteries, microporous separators for liquid lithium-ion cells and physical or chemical gels consisting of a thermoplastic (cross-linked or not) swollen by a liquid organic electrolyte. These last 10 years, we have developed several alternatives based on polyether [2–4] or polyacetals [5], providing high conductivities and good mechanical properties to be used as dry thin film electrolytes for

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lithium polymer batteries. Other teams (Watanabe [6], Ballard [7]) also proposed valid alternatives for this application. But whatever the improvements of polymer electrolytes, the best conductivities of true and pure polymer electrolytes are limited to roughly 10^{-4} S cm^{-1} at 30°C. These conductivities are too low for electronic applications and at present battery manufacturers are facing two options in lithium-ion batteries. The first one involves a microporous separator, consisting of a polyolefine crossed by a liquid organic electrolyte, while the second one is based on a thermoplastic plasticized or rather gelled by the previous liquid electrolyte. Plasticized thermoplastics, such as polyvinyl formal (PVF), and polyvinyl butyral (PVB), but also polyacrylonitrile (PAN) and vinylidene fluoride/hexafluoropropene copolymers (PVDF/HFP) originate from Feuillade [8,9] who claimed, as soon as 1973, the use of solid polymer electrolytes to shape electrochemical elements into thin films. After that, much relevant researches were carried out in this field using several materials such as PAN (Watanabe [10], Abraham [11]), PMMA (Scrosati [12]) or PVDF (Tarascon [13], Abraham [14] etc.). We selected a polynitrile matrix as a suitable thermoplastic for lithium-ion batteries, as nitrile substituents are well-known for their high dipole moment. Now the instability of nitrile groups versus metallic lithium is also well-known [15]. We identified the acidity of the hydrogen in a position of a nitrile group as one of the sources of PAN instability versus lithium. We prepared therefore, homopolymers and copolymers based on methacrylonitrile [16], in which the nitrile group is located on a tertiary carbon.

2. Experimental

2.1. Host polymer

2.1.1. Polymethacrylonitrile (polysciences)

The polymethacrylonitrile (polysciences) was used as received, lithium trifluoromethanesulphonylimide (Fluka) was dried for several days at 120°C in a dynamic vacuum and then stored in a glove box. The polymethacrylonitrile copolymers were prepared by free-radical polymerization at 70°C in bulk, using different comonomers.

2.1.2. POE-PMAN-POE block copolymers

POE-PMAN-POE block copolymers were prepared by anionic polymerization in a glove box at room temperature. An α, ω sodium dialcoholate was obtained by adding sodium hydride (Aldrich) to a solution of α, ω dihydroxyoligo(oxyethylene) (Aldrich $M_n = 2000$ – 4600 g mol^{-1}). Typically 1×10^{-5} moles of POE were dissolved in 7 ml of distilled THF. The amount of

sodium hydride was added in stoichiometry. After nearly 20 h of stirring, methacrylonitrile (Aldrich 99%, distilled on CaH_2 in a dry argon atmosphere) was added to the dialcoholate. After 24–48 h, polymerization was stopped by adding some water. The product was precipitated in pentane and dried in dynamic vacuum at 40°C.

2.2. Membrane

Different compositions of dried PMAN and copolymer electrolytes were obtained at the desired CN/Li or (CN + O)/Li ratio by dissolution under stirring of preweighted amounts of salt and polymer in dimethyl ketone.

2.3. Thermal analysis

Glass transition temperatures, T_g , and melting temperatures, T_m , were measured in helium using a Netzsch STA409 Thermal analyzer. In a typical procedure, samples were cooled rapidly from room temperature to -100°C . Then the samples were heated at $10^\circ\text{C min}^{-1}$ up to 60°C .

2.4. Gel permeation chromatography (GPC)

The GPC analyses were performed in THF, previously filtered (0.4 mm) and degassed, on a Waters 590 GPC, equipped with a differential refractometer Waters 410 and a Waters 745B data module. Four columns of 500, 10^3 , 10^4 , 10^5 Å were used with the flow rate of 1.2 ml min^{-1} .

2.5. $^1\text{H-NMR}$

The analyses were performed on a Bruker AC 200 MHz in DMSO.

2.6. Infrared spectroscopy

Mid-infrared spectra were acquired at room temperature using a NICOLET 710 FTIR system coupled to a computer. The spectra were collected over the range 400 – 4000 cm^{-1} by averaging 64 scans at a maximum resolution of 4 cm^{-1} . Samples were analyzed as thin films in transmission mode using silicium substract. The cell was made in a glove box to prevent water contamination.

2.7. Ionic conductivity

The ionic conductivities of the lithium salt/polymer complexes were determined, under vacuum for PMAN electrolytes and under argon for plasticized electrolytes by impedance spectroscopy over the frequency range 5

Hz to 13 MHz, using stainless-steel blocking electrodes and a HP4192A analyzer.

2.8. Acronyms

A block copolymer PMAN–POE–PMAN prepared from PEG2000 ($M_n = 2000 \text{ g mol}^{-1}$), with 200 MAN units added, will be referred to as the following abbreviation POE 2000/200 MAN.

3. Results and discussion

In addition to an expected improvement of the electrochemical instability in reduction, PMAN presents some advantages with respect to PAN. First of all PMAN exhibits a very good solubility in solvents of somewhat low boiling points, such as dimethyl ketone, while PAN is only soluble in high boiling point solvents, such as DMF. That may allow the incorporation of solvents such as dialkyl carbonate, often used as an electrolyte component in lithium-ion batteries. Moreover, a comparative study of the thermal degradation [17] reported that PMAN degradation leads from 220°C to a quantitative formation of methacrylonitrile by depolymerization while that of PAN results in more substantial amounts of HCN. However this advantage is not decisive. As for the drawback of PMAN as compared to PAN, we must emphasize its very high affinity for the usual solvents of lithium batteries, which prevents the formation of a chemical gel as observed with PAN. This disadvantage may be overcome by the use of PMAN copolymers. The first investigations confirm the stability improvement in reduction while the conductivities of PMAN based electrolytes are in the same range as those of PAN gelled electrolytes. Thus conductivities reached $10^{-3} \text{ S cm}^{-1}$ at 20°C and, even at 80°C, no creeping phenomenon was observed. But the electrochemical properties of ambient and sub-ambient electrolytes will be explained in detail in a following paper. Now in order to appreciate the salt/polymer interactions we investigated some solutions of lithium imide–LiTFSI—in dry PMAN and in a triblock copolymer PMAN–POE–PMAN. Ionic conductivities of several hydrated perchlorates (Li^+ , Fe^{2+} , Cr^{3+} etc.) dissolved in dry PAN were early reported by Reich et al. [18].

3.1. Conductivities in PMAN based host polymers

3.1.1. PMAN homopolymer

The viscosimetric molecular weight of PMAN determined in DMF, using the following Mark–Houwink–Sakurada relation $[\eta] = 3.06 \times 10^{-3} \times [\bar{M}_n]^{0.503}$ [19] was $M_v = 1.7 \times 10^5 \text{ g mol}^{-1}$. PMAN and lithium imide were dissolved in a common solvent, dimethyl ketone.

The viscous solution was then casted according to the classical procedures. Then dimethylketone was removed by heating under vacuum. Special attention must be paid to the solvent evaporation. Surprisingly a high amount of imide salt can be dissolved in PMAN. Although PMAN glass transition temperature is close to 120°C, the dissolution of salt, at least at medium to high salt concentration, it did not result in brittle films. On the contrary, the salt incorporation softens the material, allowing flexible membranes to be obtained for the highest salt concentrations. Nevertheless T_g 's calorimetric determination is difficult and is not able to evidence, with accuracy, the T_g decrease when the salt concentration increases. Forsyth et al. (PAN) [20,21], also dealt with salt dissolution in dry polyacrylonitrile, emphasizing the difficulties to measure T_g with accuracy. Reich et al. [18] reported a clear decrease of T_g with salt concentration but their salts were hydrated and they attributed this plasticizing effect to the water molecules. On the other hand Watanabe et al. [10] reported that in dry PAN/LiClO₄ complex, the T_g , deduced from dynamic mechanical analyses, were very close to that of the host polymer. But the alone data provided -102°C — as compared to 107°C for PAN — was obtained for a rather low salt concentration (5 mol%). The conductivities of PMAN complexes were determined for different salt compositions. The Arrhenius plots are gathered in Fig. 1. Somewhat high conductivities $10^{-4} \text{ S cm}^{-1}$ were reached at around 90°C but for an unusual high salt concentration $\text{CN/Li} = 2$, i.e. one mole of salt for two moles of nitrile repeat unit. From σ versus $1/T$ plots, the conductivity evolution seems to follow an Arrhenius type law rather than a VTF one. From these conductivity data, as well as from the complex softening, one might assume that, as opposed to POE complexes, the salt dissolution in PMAN host polymer results in a decrease of the cohesive energy density.

3.1.2. PMAN–POE–PMAN triblock copolymers

3.1.2.1. Syntheses and characterizations of the copolymers. Besides its reactivity towards free-radical initiators which allows PMAN to be prepared by thermal free-radical polymerization, methacrylonitrile may undergo a quasi-living ionic polymerization. The main advantages of living anionic polymerization lie in (i) obtaining a well-defined polymer with a control of chain length and of molecular weight distribution; (ii) in the opportunity to prepare di and sometimes triblock copolymers; (iii) in the possibility to end-cap the polymers by functional groups. Anionic homopolymerization of methacrylonitrile has already been reported [22,23] from ambient temperature to -75°C but to the best of our knowledge the synthesis of PMAN–POE–PMAN from a living central block of POE has not

been published. On the other hand attempts to prepare PAN–POE–PAN triblocks from POE dialcoholate have been made [24] but due, in particular, to the lability of the hydrogen in α position of nitrile, the attempts failed to obtain linear copolymers. In order to synthesize the triblock copolymers we have prepared, in THF solution (see Section 2) an α, ω sodium POE dialcoholate by treatment of α, ω POE diol with sodium

hydride. Then we added the appropriate amount of methacrylonitrile in order to get the suitable PMAN block length. The time reaction was determined beforehand from a preliminary kinetic study of MAN consumption, by taking and deactivating (on itaconic acid in DMSO) solution samples and analyzing them by gas chromatography (GC). Fig. 2 plots the conversion evolution with time on copolymerization performed at

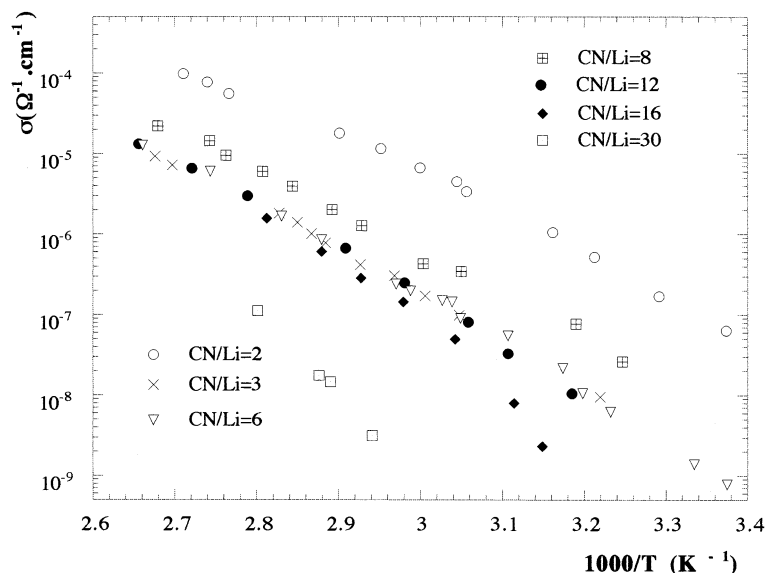


Fig. 1. Conductivities versus T^{-1} for PMAN/LiTFSI electrolytes at various salt concentrations.

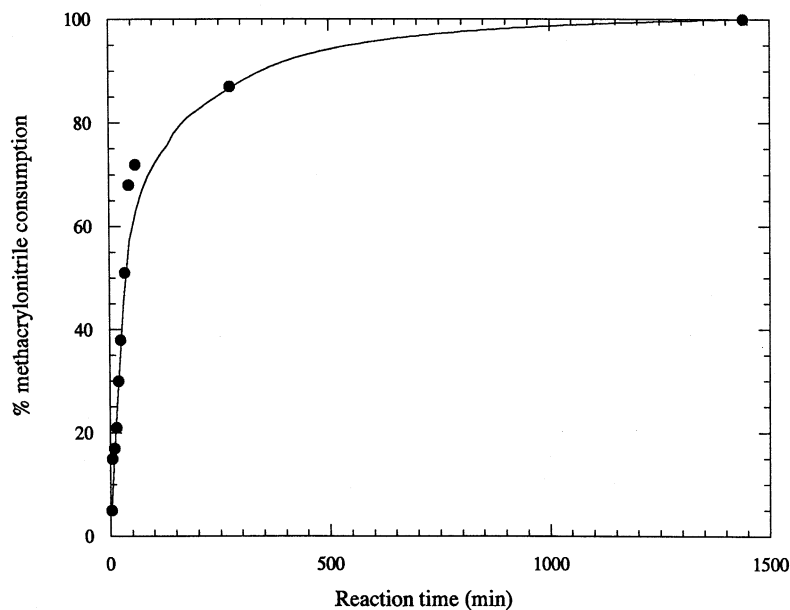


Fig. 2. Methacrylonitrile consumption, followed by GC, for the polymerization of 200 methacrylonitrile units from a starting POE (4600 g mol^{-1}) dialkoxide.

Table 1
GPC and NMR results ^a

	MAN units incorporated (NMR)	M_n (NMR) (g mol ⁻¹)	M_w (GPC) (g mol ⁻¹)	M_n (GPC) (g mol ⁻¹)	I_p
POE 2000/200 MAN	200	15 400	31 900	13 100	2.4
POE 3400/200 MAN	185	15 800	12 800	9900	1.3
POE 4600/200 MAN	199	17 900	14 300	11 400	1.3
POE 4600/200 MAN	198	17 900	18 300	11 000	1.2

^a Determination of molecular weights and compositions for some PMAN–POE–PMAN block copolymers.

Table 2
Thermal parameters, melting point, T_m , glass transition temperature, T_g and crystallinity rate for triblock PMAN–POE–PMAN copolymers

	POE 2000/200 MAN	POE 4600/87 MAN	POE 4600/200 MAN
Crystallinity rate starting POE	85%	90%	90%
Crystallinity rate copolymer	0%	15%	0%
T_m copolymer	–	20–37°C	–
T_g copolymer	–17°C	–	–20°C

ambient temperature, starting from a POE diol of 4600 g mol⁻¹. From this figure it appears that a 90% reaction rate is obtained in 6 h, while a total consumption of the monomer requires 24 h. As an excess of NaH (insoluble in THF) might initiate heterogeneous homopolymerization of MAN, we checked this opportunity in attempting the MAN homopolymerization in NaH/THF suspension. But after 24 h the PMAN yield reached only 0.05%. We must therefore be confident in our results and neglect the copolymer pollution by PMAN. The proportion of MAN units versus oxyethylene ones was determined by ¹H-NMR. Assuming a difunctional initiation of MAN and the absence of transfer/termination reactions, a theoretical M_n can be calculated for the triblock copolymers. Gel permeation chromatography, performed in THF, provided qualitative information about the absence of the starting POE, as seen from the comparison of chromatograms. In addition GPC provided the polydispersity indexes $I = M_w/M_n$ which range between 1.2 and 1.5, in agreement with a quasi-living polymerization. Lastly, using a polystyrene calibration curve, we got M_w and M_n values. Obviously these values were obtained in equivalent polystyrene but most of them were in agreement with the theoretical molecular weights calculated from NMR (Table 1).

DSC analyses were performed on several triblock copolymers. Melting points, T_m , crystallinity rates and glass transition temperatures are gathered in Table 2.

One may observe that whatever the length of the central POE block, its crystallinity vanishes. However this vanishing depends on the length of PMAN side blocks. Thus, a crystallinity rate of 15% with a T_m ranging between 20 and 37°C remained, when the average global content of MAN units was equal to 87 while no crystalline phase was seen for 200 MAN units. The DSC recordings did not allow one to evidence more than one T_g . This was in between that of pure POE and that of PMAN and close to –20°C. Although the POE central block indisputably acts as an internal plasticizer with respect to PMAN homopolymer, this DSC T_g should be checked by another method such as dynamic mechanical analysis.

X-ray diffraction confirms the amorphous character of the POE block. Lastly, FTIR spectra shows (Fig. 3), in the region 800–1400 cm⁻¹, a broadening of the polyether bands i.e. 1110–1150 cm⁻¹. Bands at 1352, 1360, 842 cm⁻¹, attributed to modes in the crystalline phase of POE [25,26], disappear with respect to the infrared spectra of the corresponding POE oligomers and confirm the amorphous character of the central POE block. On the other hand, the nitrile band at 2236 cm⁻¹ is unaffected by the copolymerization.

3.1.2.2. Ionic conductivities. All the conductivity measurements were performed on the triblock copolymer sample POE 2000/200 MAN. The salt composition was referred to as the ratio between on the one hand both

ether and nitrile solvating moieties and on the other hand lithium cation and expressed as $(O + CN)/Li$. Fig. 4 provides a comparison between some samples, corresponding to rather high salt concentration. We should emphasize that concentrations up to $(O + CN)/Li = 1$ were prepared. X-ray diffraction did not permit one to characterize, for $(O + CN)/Li = 3$ complex, any salt precipitation. The conductivities are higher than those obtained with PMAN, reaching $10^{-5} \text{ S cm}^{-1}$ at about 37°C . However, due to a creeping trend of the membrane, we did not succeed in measuring the conductivities at higher temperature. Obviously these conductivities are still too low for dry polymer electrolytes, but on the one hand these kinds of host

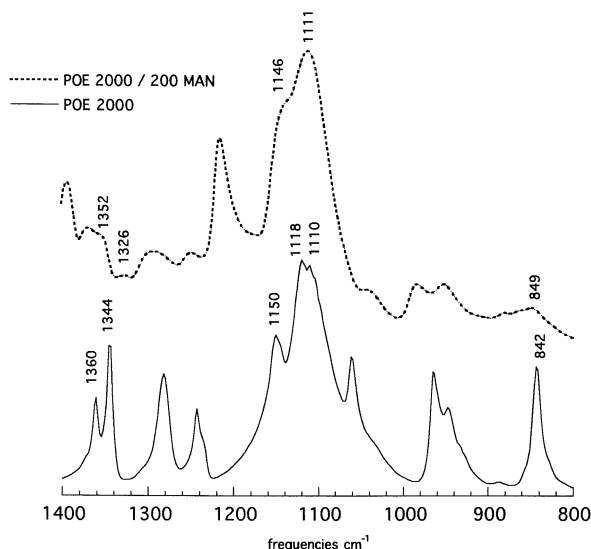


Fig. 3. Spectra of POE 2000/200 MAN and POE 2000 in the $1400\text{--}800 \text{ cm}^{-1}$ region.

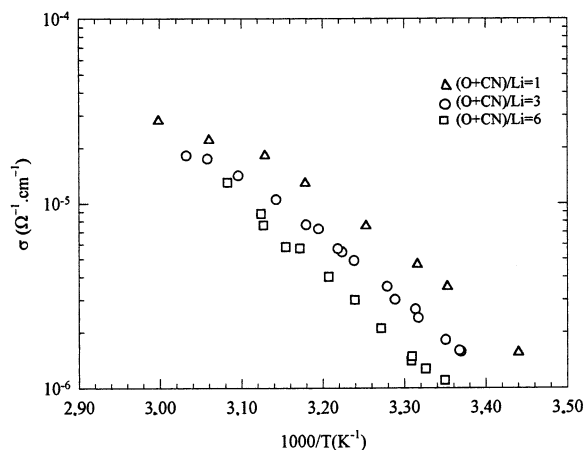


Fig. 4. Conductivity versus T^{-1} for POE 2000/200 MAN at various salt concentrations.

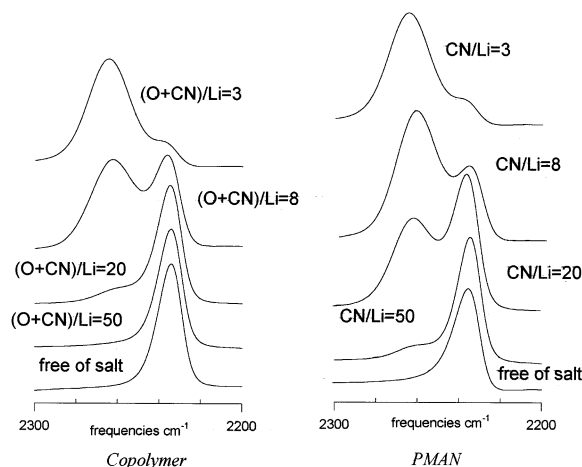


Fig. 5. Comparative evolution, with salt concentration, of nitrile in PMAN homopolymer and in POE 2000/200 MAN copolymer.

polymers are intended to be used with liquid electrolytes and on the other hand the material is far from being optimized.

3.2. Infrared spectroscopy

The central polyether block is well-known for its ability to dissolve a wide variety of salts. This ability was not identified, up to now, for polymethacrylonitrile but, owing to the donor number of nitrile groups, it might have been anticipated. FTIR spectroscopy has been widely used to analyze the interactions of polyether with lithium salts [27,28]. On the other hand, the interactions between nitrile and lithium salts have been mainly investigated through liquid solutions of LiClO_4 in acetonitrile [29] and acrylonitrile [30] or LiCF_3SO_3 in acrylonitrile [31]. As for the interactions between PAN and LiClO_4 , they were investigated through gelled electrolytes PAN/ LiClO_4 /DMF and PAN/ LiClO_4 /PC [32,33]. The solubility of PMAN in dimethyl ketone allows one to prepare PMAN/salt complexes in a wide salt concentration range. As POE and PMAN are not compatible, PMAN–POE–PMAN copolymers are well-adapted to investigate the competition between POE and PMAN in the salt solvation.

We selected the nitrile band to follow the polynitrile/salt association and to infer the proportion of salt complexed by the POE block. The complexation of lithium cations by the nitrile functions results in a shift towards higher frequencies. Thus in PMAN as well as in the triblock copolymer a shift of 25 cm^{-1} was observed, in agreement with the literature data on liquid nitriles [31]. The nitrile/lithium associated band is twice as wide as that of non-associated nitrile. This broadening was already reported by Alia et al.

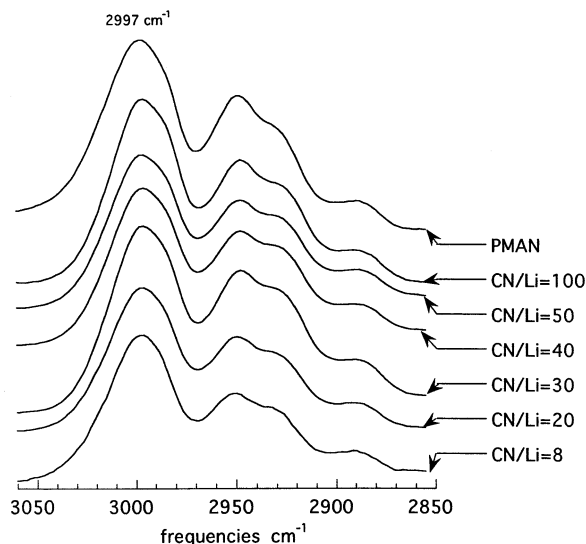


Fig. 6. Spectra of PMAN/LiTFSI complexes in the 3050–2850 cm^{-1} region.

[31]. In Fig. 5 the evolution of nitrile peaks are compared with the salt concentration, both in PMAN and in the triblock copolymer. In PMAN the nitrile associated peak is visible from the lowest salt concentration and increases regularly with the salt content. On the other hand, in the triblock copolymer, the nitrile peak remains unchanged at low salt concentration such as $(\text{O} + \text{CN})/\text{Li} = 50$. This qualitative comparison points to a higher solvating ability of POE block as compared to a PMAN one. It has been reported that the molar absorption coefficient is higher for associated nitriles than for free nitriles [31]. As a matter of fact, we observed in PMAN/salt complexes that the global intensity of the nitrile bands increases with the salt concentration. In order to determine the molar absorption coefficients B_a and B_f , respectively for the associated and free nitriles, we plotted the total intensity of nitrile bands, I_t versus the intensity of associated nitriles, I_a . To calculate I_t and I_a , we used the areas obtained by deconvolution using Peakfit software. The spectra were curve-fit to a straight base line with one Gaussian function for each band. These areas were then normalized with respect to the area of the CH_3 stretching band of PMAN at 2994 cm^{-1} . The insensitivity of the band

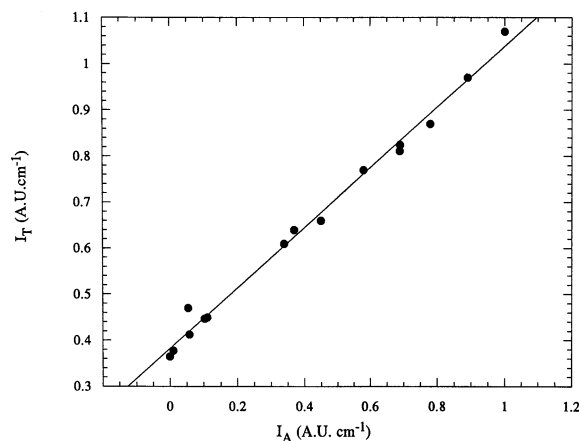


Fig. 7. Plot of the total integral intensity of the $\nu(\text{C}\equiv\text{N})$ envelope versus the integral intensity of the associated acrylonitrile component.

intensity was checked on several PMAN/LiTFSI complexes (Fig. 6). We used therefore this CH_3 band in the copolymer complexes. I_t versus I_a plot (Fig. 7) was then fitted by Irish relation already used by Alia et al. [30]: $I_t = (1 - B_f/B_a)I_a + B_fM_f$. As in PMAN, the molar concentration in methacrylonitrile is 14.9 mol kg^{-1} . B_a and B_f were determined. B_a/B_f ratio is equal to 2.91. Using this correction factor it was possible to access the fraction of associated nitrile both in PMAN and in the triblock copolymer (Table 3). From these data, it was possible to access to the evolution, with salt composition, of the solvation number in PMAN (Fig. 8). At dilute concentration, the solvation number of lithium ion in PMAN could be estimated at 4.5–5. This value is higher than those reported for lithium triflate [31] and lithium perchlorate [30] in acrylonitrile, which are equal to 3. The decrease in the solvation number, when the salt concentration is increased, can be interpreted as ion pair formation. Experiments are still in progress to confirm this behavior. Just as in PMAN, the solvation number was also determined in the triblock copolymer, assuming the absence of cross-solvation between ether and nitrile groups. Due to the low apparent solvation number of nitrile function (Table 4), determined by the fraction of associated nitrile, we can assume an heterogeneous distribution of the lithium cation. Only few

Table 3

Percentage of associated nitrile in PMAN homopolymer and in POE 2000/200 MAN copolymer ^a

	Sg/Li = 50 (%)	Sg/Li = 20 (%)	Sg/Li = 8 (%)	Sg/Li = 3 (%)
PMAN	8	29.5	52	84.5
Copolymer	0	8.5	37	73.5

^a Sg, solvating group CN or CN+O.

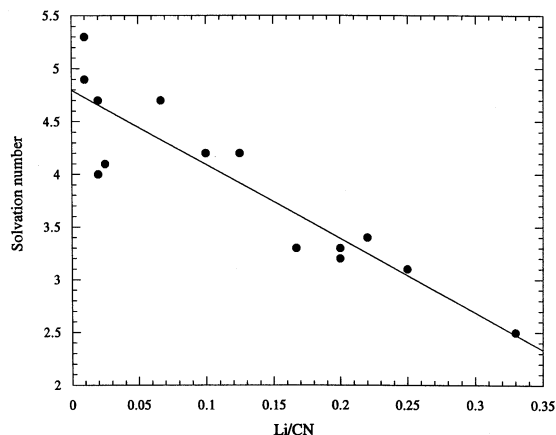


Fig. 8. Apparent solvation number of the lithium ion in PMAN, calculated from infrared results.

Table 4

Apparent solvation numbers of the lithium ion in the PMAN block and effective lithium concentrations in the POE block (POE 2000/200 MAN)^a

	Sg/Li = 50	Sg/Li = 8	Sg/Li = 3
Apparent solvation number	–	3	2.2
Effective O/Li	9.3	3.5	2

^a Sg, Solvation group CN or CN+O.

lithium cations were present in the PMAN sequence. Effective lithium concentration in POE can be calculated (Table 4), supposing that the solvation number of the nitrile function was the same in the copolymer as in the homopolymer. Even at low concentration ((O + CN)/Li = 50), relatively high salt concentration is observed in the POE sequence O/Li = 9.3. This quantitative determination confirms the prevalent solvation by POE block. Nevertheless, we must point out that the polynitrile block is able to dissolve a higher amount of salt than a POE one.

Recently, assignment of the normal vibrations of the imide anion was performed [34–37]. A preliminary study was done on the ionic association in our systems as a function of salt concentration. Bakker et al. [34] have studied ion aggregation in POE M(TFSI)₂ on the FTIR 740 cm⁻¹ absorption which was found to be particularly sensitive to ion pair formation. They assign this 740 cm⁻¹ component to $\nu(\text{S-N})$ vibration of the free imide ions, new components were observed, at high frequencies, for O/M < 9 and assigned to ion pair and aggregate. More recently, Rey et al. [36,37] proposed to assign the bands at 739 and 761 cm⁻¹, observed in FTIR, at a predominant $\delta_{\text{S}}(\text{CF}_3)$ band combined with $\nu_{\text{S}}(\text{SNS})$. In our study, the bands at 740 and 761 cm⁻¹

($\delta_{\text{S}}(\text{CF}_3) + \nu_{\text{S}}(\text{SNS})$) and 788 cm⁻¹ $\nu(\text{CS})$ were sensitive to salt concentration. Some broadening and shifts were observed in PMAN/LiTFSI (CN/Li = 50, 40, 20, 15, 8, 5, 4, 3) complexes for CN/Li ≤ 15 (Fig. 9). The ion pairs were observed at lower salt concentration than in POE complexes, this can be due to the lower solvating ability of PMAN. At dilute concentration (CN/Li ≥ 20), the peaks at 740, 761 and 788 cm⁻¹ could be attributed to a free imide anion [36,37]. For high salt concentrations, CN/Li = 3, the three bands were at 745, 767 and 798 cm⁻¹, respectively, in accordance with more associated compounds. The same behavior was observed on POE block copolymer with the ion-pairs detected for (O + CN/Li ≤ 8).

4. Conclusion

New block copolymers PMAN–POE–PMAN were prepared by anionic polymerization. These materials were investigated at different PMAN and POE block

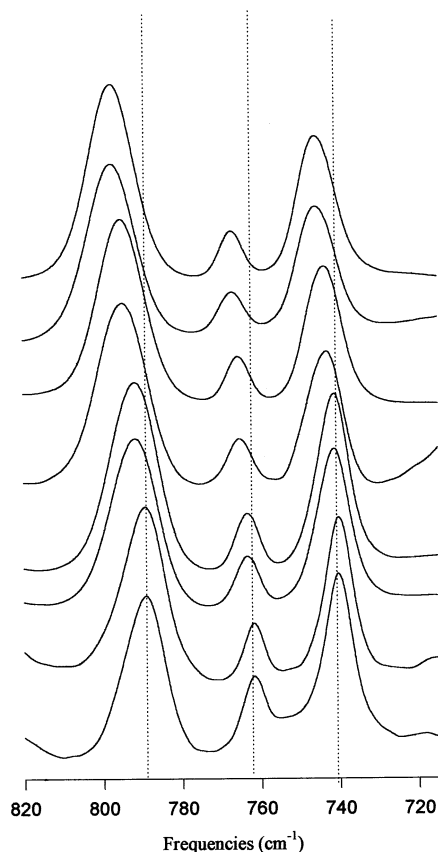


Fig. 9. Spectra of PMAN/LiTFSI complexes in the 850–700 cm⁻¹ region. Starting from dilute complex, the spectra of the successive complexes CN/Li, 50, 40, 20, 15, 8, 5, 4, 3 are shifted in ordinate for convenience.

lengths. Dried polymer electrolytes based both on these copolymers and on PMAN homopolymer exhibit significant conductivities at high salt concentration ((O + CN)/Li = 1). The FTIR comparative study of complexes based on homo and copolymers, allowed the salt/polymer interactions to be evidenced. In particular PMAN blocks appear able to dissolve a large amount of salt, while POE central block has a higher solvation ability.

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